

E. Quarrel

time limit per test: 1 second
memory limit per test: 256 megabytes

Friends Alex and Bob live in Bertown. In this town there are n crossroads, some of them are connected by bidirectional roads of equal length. Bob lives in a house at the crossroads number 1, Alex — in a house at the crossroads number n .

One day Alex and Bob had a big quarrel, and they refused to see each other. It occurred that today Bob needs to get from his house to the crossroads n and Alex needs to get from his house to the crossroads 1. And they don't want to meet at any of the crossroads, but they can meet in the middle of the street, when passing it in opposite directions. Alex and Bob asked you, as their mutual friend, to help them with this difficult task.

Find for Alex and Bob such routes with equal number of streets that the guys can follow these routes and never appear at the same crossroads at the same time. They are allowed to meet in the middle of the street when moving toward each other (see Sample 1). Among all possible routes, select such that the number of streets in it is the least possible. Until both guys reach their destinations, none of them can stay without moving.

The guys are moving simultaneously with equal speeds, i.e. it is possible that when one of them reaches some of the crossroads, the other one leaves it. For example, Alex can move from crossroad 1 to crossroad 2, while Bob moves from crossroad 2 to crossroad 3.

If the required routes don't exist, your program should output -1 .

Input

The first line contains two integers n and m ($2 \leq n \leq 500$, $1 \leq m \leq 10000$) — the amount of crossroads and the amount of roads. Each of the following m lines contains two integers — the numbers of crossroads connected by the road. It is guaranteed that no road connects a crossroads with itself and no two crossroads are connected by more than one road.

Output

If the required routes don't exist, output -1 . Otherwise, the first line should contain integer k — the length of shortest routes (the length of the route is the amount of roads in it). The next line should contain $k + 1$ integers — Bob's route, i.e. the numbers of $k + 1$ crossroads passed by Bob. The last line should contain Alex's route in the same format. If there are several optimal solutions, output any of them.

Examples

input	Copy
2 1 1 2	
output	Copy
1 1 2 2 1	

input	Copy
7 5 1 2 2 7 7 6 2 3 3 4	
output	Copy
-1	

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 29 (Div. 2, Codeforces format)

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2)

Choose file: No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
373241984	May/01/2026 18:07	Accepted
373241854	May/01/2026 18:05	Compilation error
373241617	May/01/2026 18:02	Time limit exceeded on test 29

input

Copy

7 6

1 2

2 7

7 6

2 3

3 4

1 5

output

Copy

6

1 2 3 4 3 2 7

7 6 7 2 1 5 1

373241344	May/01/2026 18:00	Accepted
373241124	May/01/2026 17:58	Time limit exceeded on test 12
373241003	May/01/2026 17:56	Compilation error
373240888	May/01/2026 17:55	Compilation error
373240804	May/01/2026 17:54	Compilation error
373240584	May/01/2026 17:52	Accepted
373240290	May/01/2026 17:49	Time limit exceeded on test 11

→ Problem tags

graphs shortest paths *2400

No tag edit access

→ Contest materials

- Announcement
- Tutorial (en)

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